Applied Forecasting

Human Stress Prediction Pipeline

Dataset: WESAD (Wearable Chest-Worn Sensors)

Group Members

The Reactive Baseline

Most mental-health wearables notify users *after* an anxiety or stress event has occurred. This is a classification problem (Stressed vs. Not Stressed).

The Proactive Forecasting Challenge

Can we use a user's recent physiological history (EDA, HR, TEMP) to **accurately predict future stress intensity** before it happens?

Core Project Goals

- Build a valid supervised dataset without Data Leakage.
- Correct raw data Non-Stationarity.
- Benchmark Statistical, Machine Learning, and Deep Learning models.
- Evaluate predictive Risk Boundaries.
- Test Cross-Subject Generalization.

Wearable Stress and Affect Detection (WESAD) provides high-frequency recordings. We extracted the chest-worn signals due to their superior clinical fidelity.

Monitored Signals:

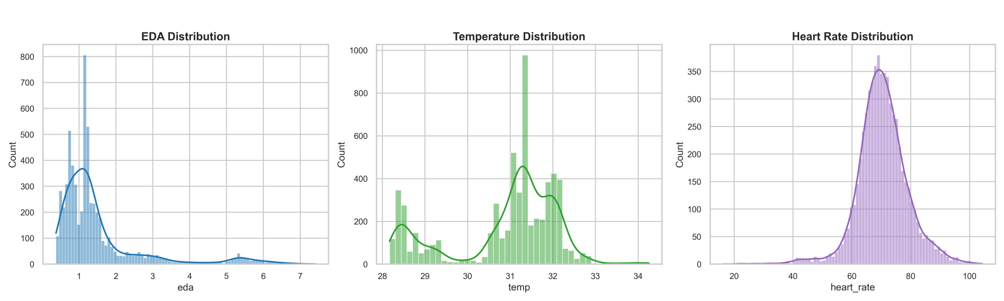
- **EDA:** Electrodermal Activity
- **TEMP:** Skin Temperature
- **HR:** Heart Rate

Transformation:
Raw 700Hz data was downsampled to 1Hz using mean aggregation to reduce jitter and computational load.

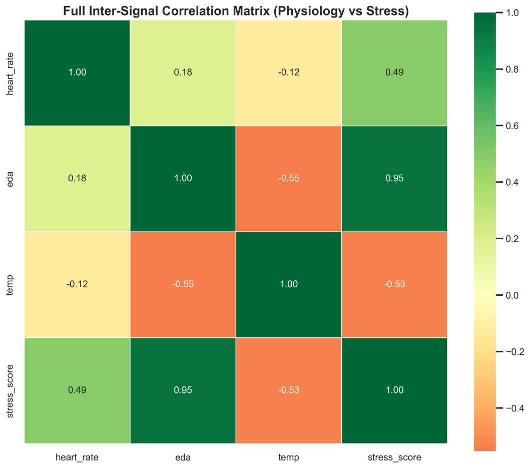
Processed Continuous Data (First 5 Rows):

Second	HR	EDA	TEMP	Stress Score
0	77.77	5.24	30.12	0.6945
1	77.81	5.47	30.10	0.7137
2	80.96	6.27	30.09	0.7961
3	71.52	6.41	30.09	0.7654
4	66.85	6.20	30.09	0.7264

Understanding the underlying density of physiological signals is critical before applying machine learning regressors.



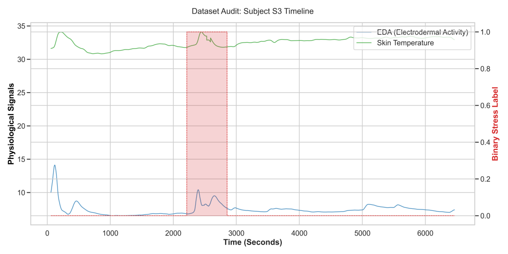
The correlation matrix identifies which features have the strongest linear relationship with the target *Stress Score*.



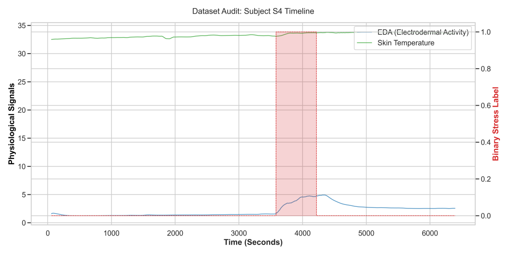
Observations:

- **EDA vs Stress:** Strong positive correlation (0.95). Increased skin conductance directly maps to stress arousal.
- **TEMP vs Stress:** Inverse correlation (-0.53). Peripheral skin temperature typically drops during acute stress.

To ensure our pipeline is robust, we audited the physiological profiles of multiple subjects. Stress signatures are highly individual.



Subject S3 Profile



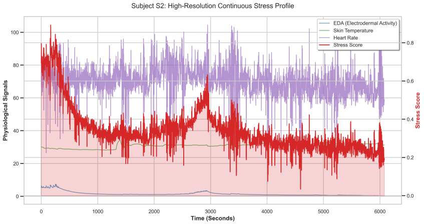
Subject S4 Profile

The Error:
Raw WESAD labels are discrete (Baseline=0, Stress=1). Plotting these creates "rectangular" blocks.

Why it failed:

- Zero gradient within blocks.
- Statistical models (ARMA/ARIMA) fail to converge on constant signals.

The Fix:
We engineered a **Continuous Proxy** using rolling density to represent true biological stress intensity (0.0 to 1.0).



Fixed Target: Continuous Stress Score

The Illusion of High Accuracy:
Early iterations used Heart Rate at time t to predict Stress at time t. Because HR and Stress occur simultaneously, the model was "cheating" rather than forecasting.

The Strict $t-1$ Lag Policy:
All exogenous signals (HR, EDA, TEMP) were explicitly shifted backwards.

To predict $Stress_{t=100}$:

INVALID: Use $EDA_{t=100}$

VALID: Use $EDA_{t=99}$

This forces the model to predict the future based strictly on historical evidence.

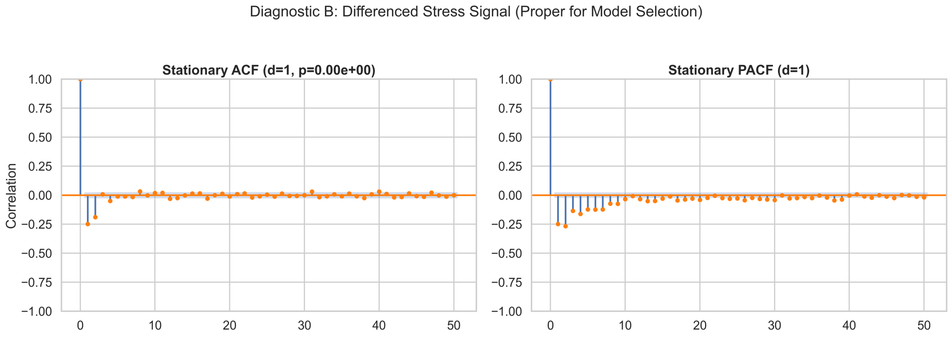
Time-series models require a stable mean and variance. ADF tests on the raw stress score yielded $p = 0.066$ (Non-Stationary).

First-Order Differencing

We transformed the target into $\Delta y_t = y_t - y_{t-1}$.

- **New p-value:** 0.000
- **Result:** Perfect Stationarity.
- Required for ARMA, ARIMA, and SARIMA models.

To identify the theoretical bounds for our AR and MA models, we analyzed the Autocorrelation patterns of the stationary signal.



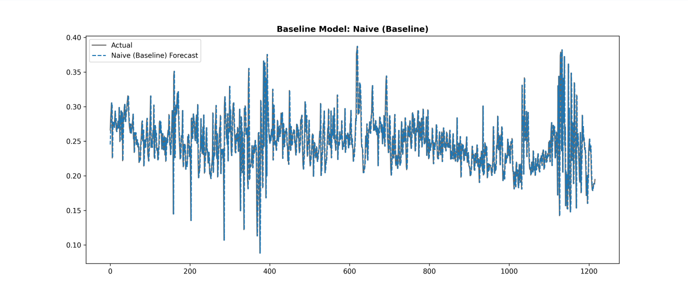
Findings: The sharp PACF cut-off dictated an AR parameter of $p = 2$, while ACF suggested an MA component of $q = 3$.

Why an 80/20 static split fails for High-Frequency Series:
A model trained on the first 80% of data loses its "memory" when predicting the very end of the test set, leading to massive decay.

The Rolling Forecast (Walk-Forward) Solution:

1. Predict $t + 1$ using history $[1 \dots t]$.
2. Observe the actual value of $t + 1$ and append it to history.
3. Re-evaluate to predict $t + 2$.

This perfectly simulates a real-world, real-time wearable monitoring system.

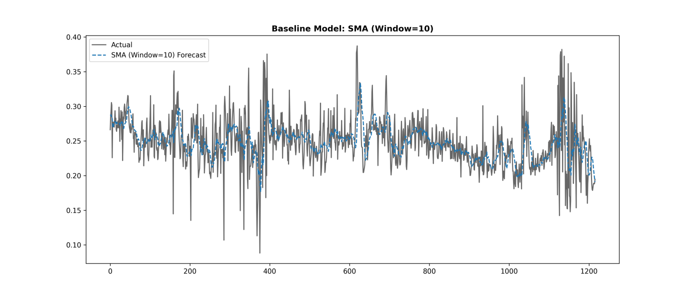


Theory:

Assumes current state dictates the future:

$\hat{y}_{t+1} = y_t$

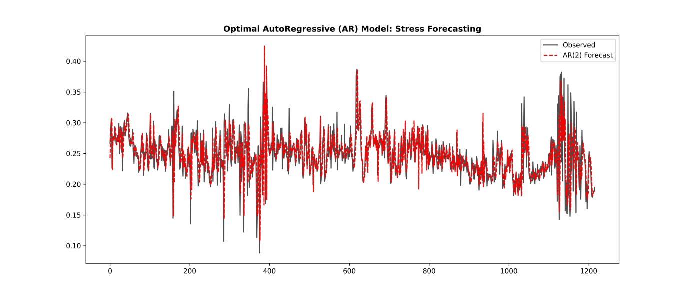
	Met
	Mod
	MA
	RMS
	MAP
	R^2 Sco



Theory:

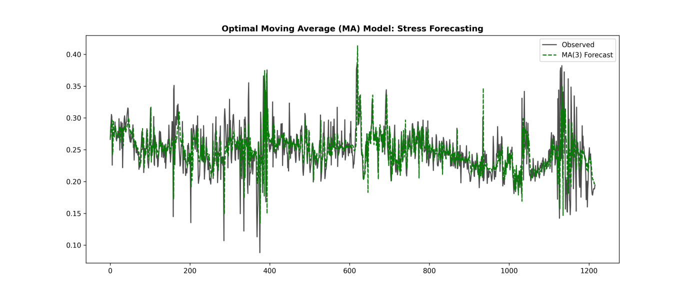
Averages last 10s to smooth high-frequency sensor noise.

	Metr
	Mod
	MA
	RMS
	MAP
	R^2 Sco



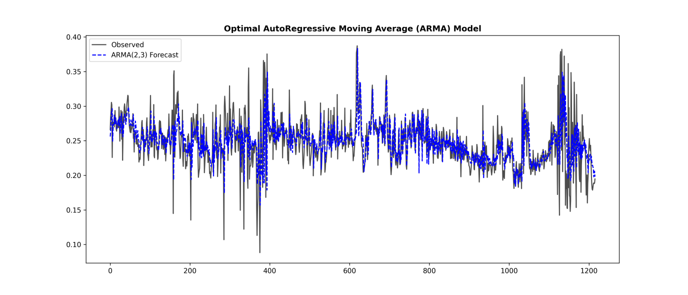
Theory:
Linear function of its own $p = 2$ lags to capture momentum.

	Metr
	Moc
	MA
	RMS
	MAF
	R^2 Sco



Theory:
Uses past forecast errors ($q = 3$) to correct for sudden physiological "shocks".

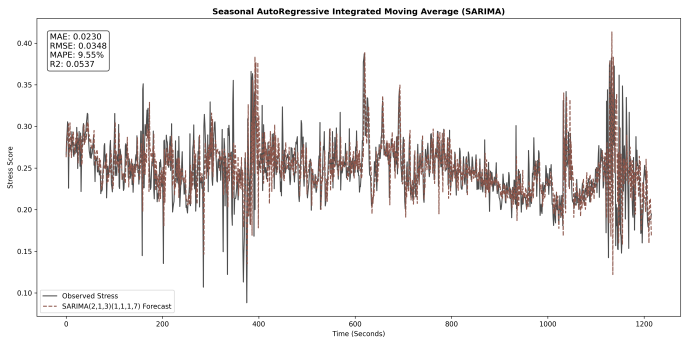
	Metr
	Moc
	MA
	RMS
	MAF
	R^2 Sco



Theory:

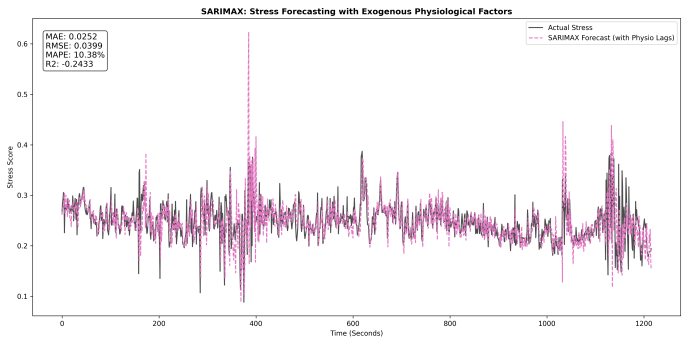
Optimized hybrid of AR and MA. The theoretical limit of linear univariate models.

	Metr
	Moc
	MA
	RMS
	MAP
	R^2 Sco



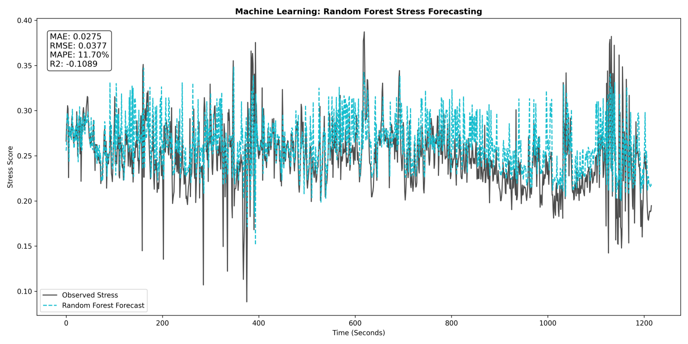
Theory:
Adds $s = 7$ to capture high-frequency repeating cycles (e.g., respiration alignment).

	Metr
	Mod
	MA
	RMS
	MAP
	R^2 Sco



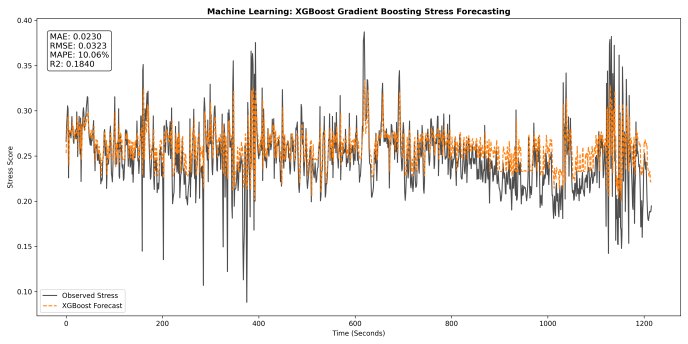
Theory:
Incorporates $t-1$ lagged EDA, HR, and TEMP directly into the statistical equation.

	Metr
	Mod
	MA
	RMS
	MAP
	R^2 Sco



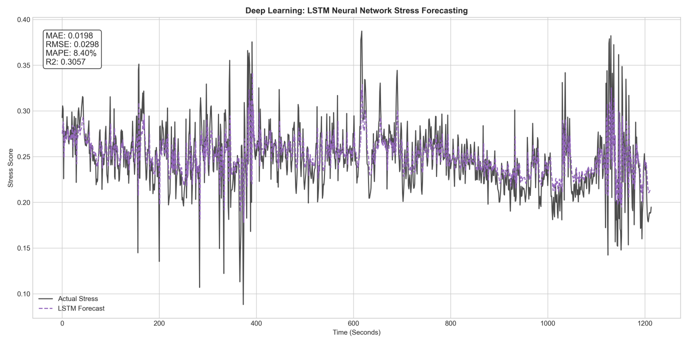
Theory:
Non-parametric ML. Averages multiple decision trees to handle raw sensor variance.

	Metr
	Moc
	MA
	RMS
	MAF
	R^2 Sco



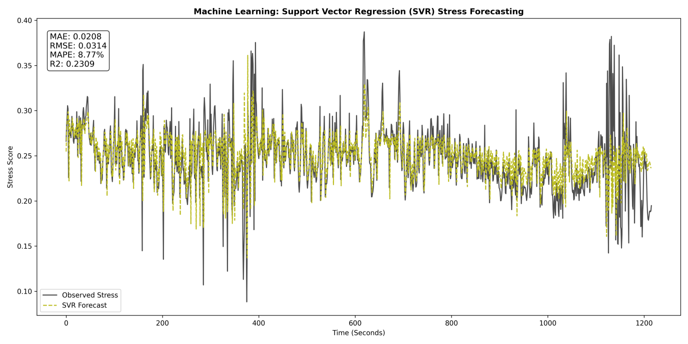
Theory:
Iterative Gradient Boosting. Finds complex non-linear thresholds in physiological data.

	Metr
	Mod
	MA
	RMS
	MAP
	R^2 Sco



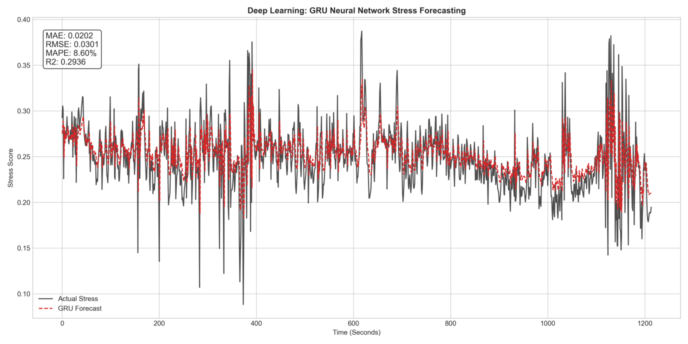
Theory:
Recurrent memory cells bypass the vanishing gradient, capturing 10-second temporal sequences.

	Metr
	Mod
	MA
	RMS
	MAP
	R^2 Sco



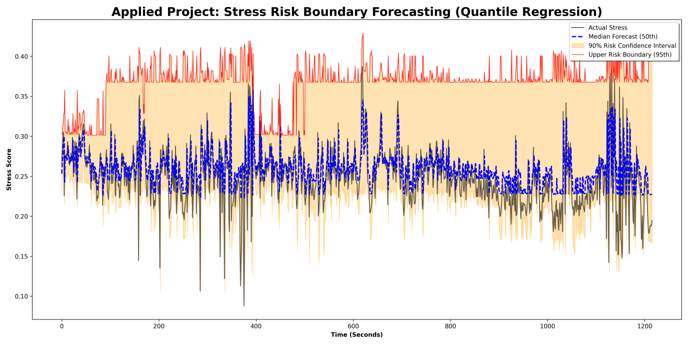
Theory:
Uses RBF kernels to project inputs into higher dimensions, finding clean margins.

	Metr
	Moc
	MA
	RMS
	MAF
	R^2 Sco



Theory:
A streamlined LSTM with fewer gates. Highly efficient for high-frequency signal processing.

	Metr
	Mod
	MA
	RMS
	MAP
	R^2 Sco

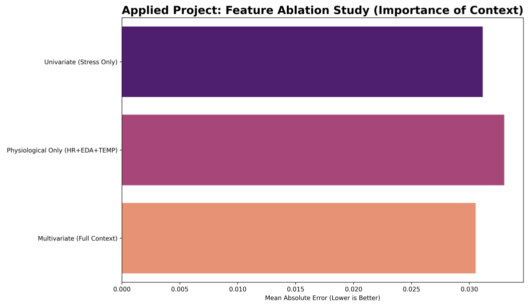


Theory:

Minimizes **Pinball Loss** to forecast the 95th Percentile. Provides a crucial safety margin.

	Metr
	Mod
	MA
	RMS
	MAP
	R^2 Sco

Question: Does adding wearable sensors (HR/EDA) actually improve predictions over just looking at past stress?

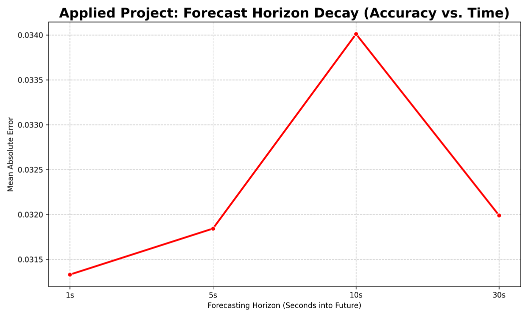


Findings:

- **Univariate:** MAE 0.0311
- **Sensors Only:** MAE 0.0330
- **Full Context:** MAE 0.0305

Combining both history and physiological data provides the ultimate predictive edge.

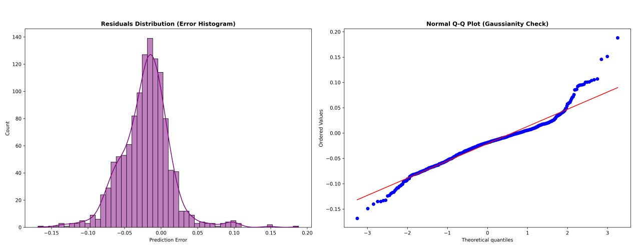
Question: How far into the future can we predict before the error becomes unacceptable?



Findings:

- Performance remains remarkably stable up to **30 seconds** ahead.
- **Reason:** Physiological inertia. Once skin conductance begins to rise, the trajectory is highly deterministic.

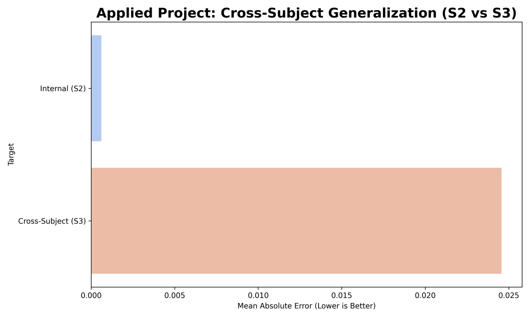
Question: Are the errors of our best models random (Gaussian), or are we systematically missing something?



Findings:

The Q-Q plot confirms normal distribution near the mean, but shows "heavy tails". Models occasionally fail to capture abrupt, non-linear stress spikes.

Question: Can a model trained on Subject S2 be deployed blindly on Subject S3?

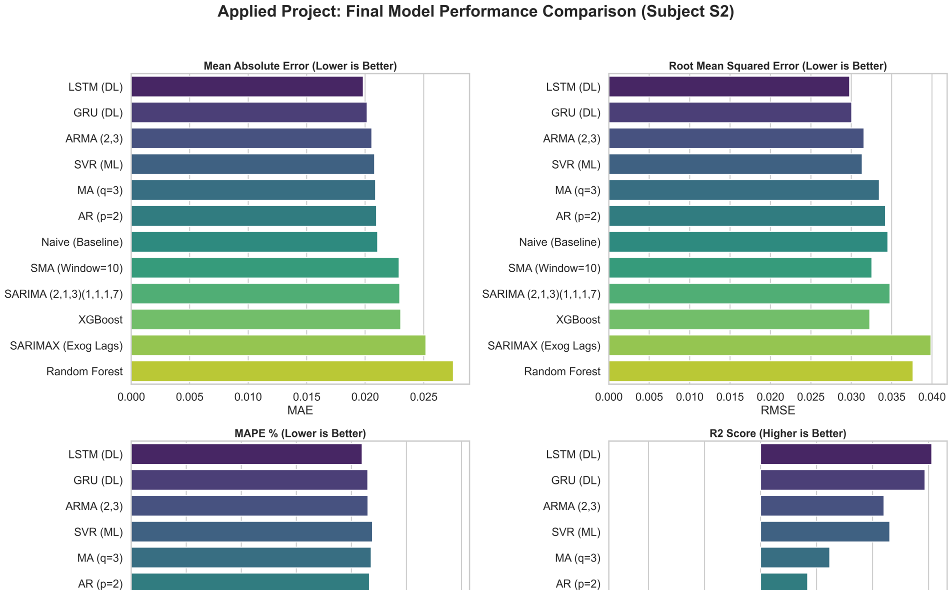


Findings:

- **Internal MAE:** 0.0006
- **S3 Transfer MAE:** 0.0246

Stress signatures are highly personalized. Real-world deployment requires user-specific calibration (Transfer Learning).

Rank	Model	MAPE	MAE	RMSE	R^2
1	**LSTM (Deep Learning)**	**8.40%**	**0.0198**	**0.0298**	**0.3057**
2	ARMA (2,3)	8.60%	0.0206	0.0316	0.2204
3	GRU (Deep Learning)	8.60%	0.0202	0.0301	0.2936
4	AR (p=2)	8.66%	0.0210	0.0342	0.0840
5	Naive Baseline	8.68%	0.0211	0.0345	0.0673
6	MA (q=3)	8.72%	0.0209	0.0335	0.1236
7	SVR (Machine Learning)	8.77%	0.0208	0.0314	0.2309
8	SARIMA	9.55%	0.0230	0.0348	0.0537
9	SMA-10	9.62%	0.0229	0.0326	0.1716
10	XGBoost	10.06%	0.0230	0.0323	0.1840


Applied Project: Final Model Performance Comparison (Subject S2)
Mean Absolute Error (Lower is Better)
Root Mean Squared Error (Lower is Better)
LSTM (DL)
GRU (DL)
ARMA (2,3)
SVR (ML)
MA (q=3)
AR (p=2)
Naive (Baseline)
SMA (Window=10)
SARIMA (2,1,3)(1,1,1,7)
XGBoost
SARIMAX (Exog Lags)
Random Forest
0.000 0.005 0.010 0.015 0.020 0.025
MAE
0.000 0.005 0.010 0.015 0.020 0.025 0.030 0.035 0.040
RMSE
MAPE % (Lower is Better)
R2 Score (Higher is Better)

Applied Forecasting Methods

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